A* Examination Guide

Built from 16 past papers and their mark schemes, January 2019 to June 2026

This guide is built from the Paper 2C question papers and mark schemes themselves. Model answers use the wording the mark schemes credit, and every warning comes from a REJECT, IGNORE, "no marks if" or "max 1" note that examiners applied. Paper 2C examines the whole specification, so Paper 1 topics still appear; this guide concentrates on what Paper 2C asks most and on the Paper 2-only content. Use it alongside the Paper 1C guide.

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Papers analysed: Jan 2019 (legacy 4CH0, 1 hour, 60 marks); Jun 2019; Jan 2020; Jan 2021; Jun 2021; Nov 2021; Jan 2022; Jun 2022; Jan 2023; Jun 2023; Nov 2023; Jun 2024; Nov 2024; Jun 2025; Nov 2025; Jun 2026. Papers are cited by the series in which they were sat, e.g. (Jun 24).

1 HOW PAPER 2C IS BUILT

Format: 1 hour 15 minutes, 70 marks, normally 6–7 structured questions (the 2019 legacy paper was 1 hour, 60 marks). That is about 1 minute per mark. Each paper opens with short recall (Periodic Table, gases, apparatus) and every later question builds to a calculation or a 3–6 mark explanation. A calculator and ruler are required.

Topic	Papers (of 16)	What is asked most
Mole calculations with gas volume (24 dm^3)	15	volume of gas from a mass; tonnes to dm^3 in standard form; % yield
Electrolysis	14	half-equations, oxidation/reduction statements, products, observations
Carboxylic acids and esters	14	naming, displayed ester, acid + carbonate/metal, ester conditions
Titration (method and calculation)	11	apparatus, indicator colours, concordant mean, concentration
Alcohols (fermentation, hydration, oxidation)	11	conditions, equations, compare the two methods, dichromate
Salts and solubility curves	11	precipitation method, crystallisation, reading and scaling curves
Metallic bonding and alloys	11	layers slide, delocalised electrons, why alloys are harder
Energy diagrams (level / profile)	10	correct arrows for ΔH and Ea; products below reactants
Bond energy calculations	9	ΔH with sign; find an unknown bond energy

Topic	Papers (of 16)	What is asked most
Equilibrium and yield	9	dynamic equilibrium; effect of temperature, pressure, catalyst
Groups 1 and 7	9	reactivity explanations, observations, displacement
Condensation polymers	8	polyester repeat unit, water as product, biodegradable
Ionic compounds and conduction	8	solid vs molten vs solution; dot-and-cross
Crude oil and cracking	7	fractional distillation description, conditions, reasons
Isotopes and A_r	7	definition, A_r to 1 d.p., isotope probabilities

The recurring long questions (3–6 marks)

- **Titration method or faults:** Jan 20 (4), Jan 23 (6), Jun 26 (3 + 2). **Standard solution:** Jun 26 (6).
- **Fractional distillation description:** Jan 21, Jan 23, Jun 24, Nov 24 (3–4 each).
- **Fermentation versus hydration:** Jun 21 (6), Jun 23 (4).
- **Metals and alloys:** Jun 19 (3), Jun 22 (3), Nov 23 (3), Nov 25 (6, iron vs high-carbon steel).
- **Why one substance conducts and another does not:** Jun 22 (4), Jun 24 (3 and 4), Jun 25 (5).
- **Group reactivity explanation:** Jan 21, Jan 22, Jun 22, Jun 24, Nov 24 (3–4 each).
- **Halogen displacement:** Jun 19 (4), Nov 23 (3), Nov 25 (5).
- **Salt preparation:** Jan 20 (5), Jan 21 (4), Jan 23 (3), Jun 24 (3), Nov 24 (5), Jun 25 (4).
- **Comparing catalysts (method):** Jan 22 (5), Nov 25 (5). **Yield and rate together:** Jun 26 (5).

2 TWENTY MARK-SCHEME RULES FOR PAPER 2C

1. **Dynamic equilibrium:** the rate of the forward reaction equals the rate of the backward reaction, and the concentrations of reactants and products remain constant. “Concentrations are equal/the same” is rejected.
2. **Yield questions:** state the effect (increases/decreases/no effect), then the reason: “the forward reaction is exothermic” or “fewer moles of gas on the left”. Quoting Le Chatelier’s principle alone is ignored. “Fewer atoms” is rejected; say moles or molecules of gas.
3. **Catalyst and equilibrium:** no effect on yield, because it increases the rate of the forward and backward reactions equally.
4. **Catalyst definition:** “provides an alternative pathway with a lower activation energy”. Jun 26 made the lower-Ea mark depend on the pathway mark and ignored “lowers the activation energy” on its own.
5. **Oxidation at the anode:** “chloride ions lose electrons”. “Chlorine loses electrons” is rejected. The same applies to bromide (Jun 24) and to reduction: “Cu^{2+} ions gain electrons”, not “copper gains electrons”.
6. **Ionic compounds conduct because ions move.** Any mention of electrons moving in an ionic compound scores 0 for that part.
7. **Metals conduct because delocalised electrons move.** Any mention of ions moving in a metal scores 0. “Free electrons” and “sea of electrons” alone are ignored; write “delocalised”.
8. **Malleability:** layers of atoms (or positive ions) slide over one another. Molecules, intermolecular forces, ionic or covalent bonding in a metal cost the marks.
9. **Alloys:** different-sized atoms disrupt the regular layers, so the layers cannot slide as easily. An alloy is a mixture; calling it a compound is rejected.
10. **Bond energies:** ΔH = energy to break bonds − energy released making bonds. The sign is required; an unsigned answer scores 2 of 3.
11. **Exothermic in bond terms:** more energy is released making bonds than is taken in breaking bonds. Saying energy is needed to make bonds, or released when breaking bonds, scores 0.

12. **Arrows on energy diagrams:** ΔH runs from the reactant level to the product level (down for exothermic). Ea points up from reactants to the peak. Jun 25 rejected double-headed arrows and plain lines.
13. **Titration readings to 2 d.p.:** 23.80, not 23.8. The mean uses concordant titres only (within 0.20 cm^3) and is given to 2 d.p., e.g. 30.30. "30.3" scored 1 of 2.
14. **Universal indicator is rejected** for titrations: it has no sharp colour change.
15. **Colour changes in the right order:** dichromate orange → green; methyl orange yellow → red when acid is added to alkali. Reversed colours score 1 of 2 at best.
16. **Molar volume:** 24 dm^3 = 24 000 cm^3. Check which unit the answer line asks for, and give standard form when asked (4.8×10^6 dm^3).
17. **Displayed formulae show every bond,** including O–H. Writing –OH in a displayed formula is rejected.
18. **Copper deposit:** pink-brown solid (or coating). "Red", "orange" and "precipitate" have each been rejected. Magnesium oxide is a "white powder/ash", not a "white precipitate".
19. **Group 1 explanation:** "the outer electron is more easily lost" (singular). Writing "electrons" in the final step has cost a mark (Jun 24, Nov 24).
20. **Never round to 1 significant figure** unless asked. Give 3 s.f. by default and keep 3+ s.f. in working.

3 Atoms, isotopes and the Periodic Table

MODEL ANSWER · ISOTOPES (2) · NOV 24

Same number of protons (same atomic number), so they are atoms of the same element. They have the same chemical properties because they have the **same electron configuration**. "Same number of electrons in the outer shell" was ignored, and any mention of neutrons earns nothing in the second mark.

- Isotopes differ in the number of neutrons (Rb-85 has 48, Rb-87 has 50). A species with three more protons than electrons is a 3+ ion (Jan 23).
- Noble gases are unreactive because they have a full outer shell (8 electrons, e.g. 2.8), so they do not need to lose, gain or share electrons.
- Electron configurations asked: 2.8.3 (Al), 2.8.4 (Si), 2.8.5 (P). Na_2S from Na^+ and S^{2-}.

Calculation	Working	Answer
A_r of Rb (Jan 19)	(85 × 72.2 + 87 × 27.8) ÷ 100 = 85.556	85.6 (1 d.p.)
A_r of Cl (Jun 19)	(35 × 77.78 + 37 × 22.22) ÷ 100 = 35.444	35.4
A_r of Mg (Jun 25)	(24 × 76.5 + 25 × 10.5 + 26 × 13.0) ÷ 100 = 24.365	24.4
A_r of Br (Nov 25)	(79 × 42 + 81 × 58) ÷ 100 = 80.16	80.2
Mass of a Li-7 nucleus (Nov 24)	$3 \times 1.6726 \times 10^{-24} + 4 \times 1.6740 \times 10^{-24}$	1.1714×10^{-23} g
Cl_2 molecules, 75% ^{35}Cl (Nov 25)	P(70) = 0.75 × 0.75; P(72) = 0.75 × 0.25 × 2 (either atom can be 37)	0.5625; 0.375

Unrounded answers (35.444 when 1 d.p. was asked) score 2 of 3. In the probability question the × 2 carried its own mark: explain that either chlorine atom can be the heavier isotope.

4 Groups 1 and 7

Group 1 with water

- Observations (any 2–3): effervescence/fizzing; moves on the surface (floats); gets smaller and disappears; sodium melts into a ball; a white trail. Potassium: lilac flame and faster. "Gas given off" is ignored. For lithium, any mention of a flame limited the answer to 1 mark (Nov 24).
- $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$. Universal indicator turns blue/purple because hydroxide ions, OH^-, are formed. Phenolphthalein turns pink. $K_2O + H_2O \rightarrow 2KOH$.
- Calcium + water → calcium hydroxide + hydrogen ($Ca + 2H_2O \rightarrow Ca(OH)_2 + H_2$): fizzing, calcium disappears, tube feels warm.
- Most reactive in Group 1 is francium; most reactive in Group 7 is fluorine (Nov 21).

MODEL ANSWER · WHY POTASSIUM/SODIUM IS MORE REACTIVE THAN LITHIUM (3) · JAN 21, JAN 22, JUN 24, NOV 24

The atom is larger: the outer shell is further from the nucleus (more shells; Li 2.1, Na 2.8.1, K 2.8.8.1). So the outer electron is less strongly attracted to the nucleus, so **the outer electron is more easily lost**.

Group 7

Halogen	State at 25 °C	Colour (as credited)
fluorine	gas	pale yellow
chlorine	gas	(pale) green
bromine	liquid	brown (red-brown); orange in solution
iodine	solid	dark grey / grey-black
astatine	solid (predicted)	black

MODEL ANSWER · FLUORINE IS MORE REACTIVE THAN CHLORINE (4) · JUN 22

Fluorine is more reactive (reactivity decreases down the group). A fluorine atom is smaller: its outer shell is closer to the nucleus. So there is a stronger attraction from the nucleus for an incoming electron. So fluorine **gains (accepts) an electron more readily**.

MODEL ANSWER · DISPLACEMENT SHOWS THE ORDER OF REACTIVITY (5) · NOV 25, ALSO JUN 19, NOV 23

Reactivity decreases down the group. A more reactive halogen displaces a less reactive halogen from a solution of its halide ions. Example: chlorine added to potassium bromide solution turns it orange, because bromine is displaced: $Cl_2 + 2KBr \rightarrow 2KCl + Br_2$. Iodine added to potassium chloride solution gives no change, because iodine is less reactive than chlorine. Order: chlorine > bromine > iodine.

Nov 23 (3 marks): chlorine reacts with bromide and iodide; bromine reacts only with iodide; iodine reacts with neither. One mark is deducted once for mixing up -ide and -ine.

- $2Li + Cl_2 \rightarrow 2LiCl$ (no charges on Li or Cl_2 in the equation).
- Testing lithium chloride (5, Jun 22): flame test → red (crimson) flame ("brick red" and "orange-red" rejected); add dilute nitric acid then silver nitrate solution → white precipitate. Using HCl or H_2SO_4 loses the acid mark.

5 AIR, POLLUTION AND RUSTING

Question	Working	Answer
% O_2 by copper in syringe (Jun 23)	volume used = 100 − 27 = 73; total = 280 + 100 = 380; 73 ÷ 380 × 100	19% (using 280 alone scores 3)

Question	Working	Answer
Air left after 2000 cm^3 burns Mg (Nov 23)	79% of air remains: 2000 × 79 ÷ 100	1580 cm^3 (1600 also credited)

- Magnesium burning: bright white light/flame and a white powder (ash). $2Mg + O_2 \rightarrow 2MgO$. The water level rises because oxygen is used, so the volume/pressure of gas falls and water takes its place (Nov 23).
- Thermal decomposition: $CuCO_3 \rightarrow CuO + CO_2$, green → black (Jun 23). $2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$; heat to constant mass so that all of it has decomposed (Jan 19).
- Reactions that produce CO_2: complete combustion of an alkane; thermal decomposition of a carbonate. Not cracking, metal + acid or rusting (Jan 22).
- CO_2: greenhouse gas, so global warming/climate change (melting ice caps, flooding, ocean acidity). Acid rain is ignored and any ozone-layer link is rejected.
- Nitrogen oxides: nitrogen and oxygen from the air react at the high temperature in an engine. Any suggestion that the nitrogen comes from the fuel is rejected. Effects: acid rain (nitric acid), respiratory problems. Catalytic converter: $2NO + 2CO \rightarrow N_2 + 2CO_2$ (Jun 19).
- Carbon monoxide reduces the capacity of the blood to carry oxygen (binds to haemoglobin).

MODEL ANSWER · RUSTING EXPERIMENT WITH TWO CONTROL TUBES (4) · JUN 21

Tube with boiled water under oil: air/oxygen is needed for rusting; boiling removes dissolved air and the oil keeps air out. Tube with a drying agent: water is needed for rusting; the drying agent removes moisture.

Rust is (hydrated) iron(III) oxide; “iron oxide” with a wrong oxidation state is rejected. Rust colour: red-brown/orange-brown.

MODEL ANSWERS · PAINT (2) AND SACRIFICIAL PROTECTION (2) · JUN 24, NOV 25

Paint forms a barrier that stops water and oxygen reaching the iron.

A more reactive metal (zinc, magnesium) is attached to or coated on the iron, so it oxidises/corrodes instead of the iron. “Rusts instead of iron” is rejected, and “sacrificial protection” is rejected as a barrier method.

6 IONIC AND COVALENT BONDING

DEFINITIONS

Covalent bond (2): electrostatic attraction between a shared pair of electrons and the nuclei of the bonded atoms. “Nucleus” (singular) loses the second mark (Jun 24, Jun 26).

Ionic formation, MgO (2): magnesium loses two electrons (2.8.2 → 2.8); oxygen gains two electrons (2.6 → 2.8). Dot-and-cross (3, Jun 25): Mg^{2+} as 2.8 and O^{2-} as 2.8 with charges. K_2O (Nov 21): two K^+ ions and one O^{2-} ion; showing only outer shells caps the diagram at 2.

MODEL ANSWER · HIGH MELTING POINT OF A GIANT IONIC COMPOUND (4) · JAN 19 (SODIUM FLUORIDE)

Giant ionic lattice; strong electrostatic forces of attraction; between oppositely charged ions; a large amount of energy is needed to overcome them. Any reference to covalent bonds, metallic bonding or intermolecular forces limits the answer to 1 mark. “More energy” is ignored.

- Simple molecular boiling point (3, Jun 21): $C_{12}H_{26}$ has larger molecules, so stronger intermolecular forces, so more energy is needed to overcome them. Implying covalent bonds break scores 0 for M2 and M3.
- Covalent compounds do not conduct: there are no electrons (or ions) free to move (Jun 26). Diamond has no delocalised electrons.
- Dot-and-cross for hydrazine, N_2H_4 (Jan 23): one shared pair N–N, two shared pairs from each N to H, and one lone pair on each nitrogen.

MODEL ANSWER · HCL IN WATER VERSUS IN AN ORGANIC SOLVENT, DRY BLUE LITMUS (4) · JUN 24

In the organic solvent the litmus stays blue, because HCl stays as molecules and no H^+ ions form (it does not dissociate). In water the litmus turns red, because HCl forms H^+ ions (hydrochloric acid). Saying the litmus is bleached loses the mark.

7 METALLIC BONDING AND ALLOYS

DEFINITIONS

Metallic bonding (MCQ, Jun 19): electrostatic attraction between positive ions and delocalised electrons.

Alloy (Jun 19): a mixture of a metal with one or more other elements. "Compound" or any chemical bonding is rejected.

MODEL ANSWERS · CONDUCTS (2) AND MALLEABLE (2) · JAN 20, JAN 21, JAN 22, JAN 23, NOV 24

Conducts: the electrons are delocalised, and they can move/flow through the structure.

Malleable: the atoms (positive ions) are in layers, which can slide over one another. For a 3-mark diagram (Nov 21): a 2-D diagram with at least two layers, particles labelled positive ions or atoms, and a statement that the layers slide.

MODEL ANSWER · WHY AN ALLOY IS HARDER THAN THE PURE METAL (3) · JUN 19, JUN 22, NOV 23

In the pure metal all the atoms are the same size and arranged in regular layers that slide easily. In the alloy the atoms are of different sizes (e.g. magnesium atoms are larger than aluminium atoms), which disrupts/distorts the regular arrangement. So it is more difficult for the layers to slide over one another.

MODEL ANSWER · PURE IRON VERSUS HIGH-CARBON STEEL (6) · NOV 25

High-carbon steel is an alloy. Pure iron is more malleable, so it is used for gates, railings, nails, wires and car bodies. Steel is harder, so it is used for cutting tools, blades, springs, hammers and rail tracks. Steel contains carbon atoms of a different size, which disrupt the regular arrangement, so the layers of iron atoms cannot slide over each other. Mentioning intermolecular forces, ionic or covalent bonds caps the answer at 5.

Use of aluminium (Nov 23)	Property to pair with it
drink cans; foil	malleable; does not corrode; non-toxic
aircraft, bicycle frames, car bodies	low density; does not corrode
overhead power cables	good conductor of electricity; low density; ductile
saucepans (Jun 19)	good conductor of heat; does not react with food; high melting point

Each use earns a mark only with a matching property, so give pairs. "Light" and "non-toxic" on their own were ignored for the pan question.

8 CONDUCTIVITY AND ELECTROLYSIS

MODEL ANSWER · METAL CONDUCTS, IONIC SOLID DOES NOT, MOLTEN/SOLUTION DOES (4–5) · JUN 22, JUN 24, JUN 25

Magnesium has delocalised electrons that can move. Magnesium chloride is ionic: in the solid the ions are in fixed positions in a lattice and cannot move, so it does not conduct. When molten or in aqueous solution the ions are free to move, so it conducts.

Water (Jun 25): covalent molecules, so not enough ions and no delocalised electrons free to move. Any mention of intermolecular forces in magnesium chloride loses a mark.

Electrode	Charge	Attracts	Process	Product rule
cathode	negative	cations (positive ions)	reduction: gain of electrons	metal if molten; in solution hydrogen unless the metal is below hydrogen (Cu, Ag)
anode	positive	anions (negative ions)	oxidation: loss of electrons	halogen from a halide; otherwise oxygen (from water or OH^-)

At the cathode	At the anode
$Cu^{2+} + 2e^- \rightarrow Cu$	$2Cl^- \rightarrow Cl_2 + 2e^-$
$2H^+ + 2e^- \rightarrow H_2$	$2Br^- \rightarrow Br_2 + 2e^-$
$Na^+ + e^- \rightarrow Na$; $Mg^{2+} + 2e^- \rightarrow Mg$	$2O^{2-} \rightarrow O_2 + 4e^-$ (aluminium oxide)
$Pb^{2+} + 2e^- \rightarrow Pb$; $Zn^{2+} + 2e^- \rightarrow Zn$	$4OH^- \rightarrow O_2 + 2H_2O + 4e^-$
$Al^{3+} + 3e^- \rightarrow Al$	$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$

MODEL ANSWERS · HOW A PRODUCT FORMS AT AN ELECTRODE (3) · NOV 23, JUN 23, JUN 26

Hydrogen at the cathode in aqueous NaCl: the solution contains hydrogen ions from water; H^+ ions are attracted to the negative electrode; they gain electrons and combine in pairs to form hydrogen molecules ($2H^+ + 2e^- \rightarrow H_2$). Sodium is not formed because sodium is more reactive than hydrogen.

Chlorine at the anode: chloride ions are attracted to the positive electrode; they lose electrons to form chlorine atoms; these combine in pairs to form chlorine molecules ($2Cl^- \rightarrow Cl_2 + 2e^-$).

Copper at the cathode: Cu^{2+} ions are positive, are attracted to the negative electrode, accept electrons and become copper atoms ($Cu^{2+} + 2e^- \rightarrow Cu$).

Observations and explanations examiners credit

- Copper(II) sulfate solution: pink-brown solid on the cathode; bubbles (oxygen) at the anode. The blue colour fades because Cu^{2+} ions, which cause the colour, are discharged/removed from the solution (Jan 20).
- Mass of cathode against time (Jan 22): rises because copper is deposited; becomes horizontal when no copper(II) ions are left in the solution.
- Molten lead(II) bromide (Jun 24): brown vapour at the anode; graphite electrodes because they conduct, have a high melting point and are inert. Electrons flow through the wires (Jun 26), not ions.
- Brine: hydrogen gives a squeaky pop with a lighted splint (a glowing splint is rejected); chlorine turns damp litmus white (bleached). Less chlorine is collected than expected because some dissolves. Chlorine is toxic, so use a fume cupboard.
- Zinc sulfate (Nov 25): near the anode OH^- ions are removed (H^+ ions form), so universal indicator turns red. Zn^{2+} ions gain electrons.
- Collecting a gas: fill a test tube with the solution, invert it over the electrode and collect by displacement (Jan 22).

9 EXTRACTION AND USES OF METALS

- An **ore** is a rock containing a metal compound ("mineral" is ignored). Unreactive metals such as gold and platinum are found uncombined.
- Metals above carbon (K, Na, Ca, Mg, Al) are extracted by **electrolysis**; metals below carbon (Zn, Fe, Pb) by **heating with carbon**. Give the method and the reason: "carbon is more reactive than lead, so carbon can displace lead" (Nov 21). Naming electrolysis for iron scores 0 (Nov 24).
- Iron(III) oxide is reduced because it loses oxygen (Jan 21).

MODEL ANSWERS · ALUMINIUM BY ELECTROLYSIS (JAN 21)

Aluminium cannot be extracted with carbon because aluminium is more reactive than carbon.

At the cathode, Al^{3+} ions are attracted to the negative electrode, where they gain electrons ($Al^{3+} + 3e^- \rightarrow Al$). At the anode, $2O^{2-} \rightarrow O_2 + 4e^-$.

The anodes must be replaced because they are made of carbon, which reacts with (burns in) the oxygen produced.

- Titanium (Jun 22): $TiCl_4$ is reduced by magnesium under argon. Argon is unreactive, so it will not react with/oxidise the magnesium or titanium, whereas oxygen in air would.
- Choosing a metal from data (Jan 19): the metal that is liquid at the operating temperature is the one whose melting point is below that temperature. Quote the numbers.
- Mass of metal from its oxide (Nov 21): 1275 g Al_2O_3 ÷ 102 = 12.5 mol → 25 mol Al × 27 = 675 g.

10 MOLES, GAS VOLUMES AND TITRATION CALCULATIONS

THE THREE FORMULAE

moles = mass ÷ M_r · moles of gas = volume in dm^3 ÷ 24 = volume in cm^3 ÷ 24 000 · moles in solution = concentration (mol/dm^3) × volume (cm^3) ÷ 1000

Tonnes: 1 tonne = 10^6 g. Answers in tonne-scale questions are expected in standard form.

Type	Past-paper example and working	Answer
mass → gas volume	Zn: 16 ÷ 65 = 0.246 mol H_2 × 24 (Nov 25)	5.9 dm^3
gas volume → mass	H_2: 550 ÷ 24 000 = 0.0229; Li = × 2 = 0.0458 × 7 (Nov 24)	0.321 g
total gas from a decomposition	$2Mg(NO_3)_2 \rightarrow 2MgO + 4NO_2 + O_2$: 7.7 ÷ 148 = 0.052; × 5/2 = 0.13 × 24 (Jan 20)	3.1 dm^3 (2 s.f.)
tonnes → gas, standard form	23.4 t NaCl: 2.34 × 10^7 ÷ 58.5 = 4.0 × 10^5; Cl_2 = ÷ 2; × 24 (Jun 21)	4.8 × 10^6 dm^3
	20 t TiO_2 ÷ 80 = 2.5 × 10^5; Cl_2 = × 2; × 24 (Jun 22)	1.2 × 10^7 dm^3
	6.6 t H_2 ÷ 2 = 3.3 × 10^6; CH_4 = ÷ 3; × 24 (Jun 23)	2.6 × 10^7 dm^3
gas with % yield	50.8 g I_2 ÷ 254 = 0.200; HI = × 2 × 80/100 = 0.32 × 24 000 (Jun 25)	7.68 × 10^3 cm^3
find the molar volume	0.150 g Li ÷ 7 = 0.0214; H_2 = 0.0107; 254 cm^3 ÷ 0.0107 (Jan 22)	23 700 cm^3/mol
gas → concentration	1570 cm^3 N_2 ÷ 24 000 = 0.0654; NH_3 = × 4; ÷ 1.100 dm^3 (Jan 23)	0.238 mol/dm^3 (3 s.f. required)
gas → solution	60 cm^3 Cl_2 = 0.0025 mol = NaClO; in 4.00 cm^3 (Jun 19)	0.625 mol/dm^3
multi-step real data	$Na_2O_2 + CO_2$: 480 dm^3 × 140 ÷ 24 = 2800 mol × 78 g (Jan 19)	218 kg
	206 dm^3 CO_2 over 4 km = 51.5 dm^3/km ÷ 24 = 2.15 mol × 44 (Jan 21)	94.4 g/km
precipitate mass and % yield	0.0125 mol $CuCl_2$ → 0.0250 mol AgCl × 143.5; 0.744 ÷ 0.850 × 100 (Jan 21)	3.59 g; 87.5%
	0.050 mol bromide → 0.025 mol $PbBr_2$ × 367 (Jun 24)	9.18 g
energy from moles	0.55 kg NaOH = 550 ÷ 40 = 13.75 mol × 47 kJ (Jun 26)	646 kJ
empirical formula	C 40.0, H 6.7, O 53.3 → 3.33 : 6.7 : 3.33 (Jun 22)	CH_2O → CH_3COOH
	C 60.0, H 13.3, O 26.7 → 5.0 : 13.3 : 1.67 = 3 : 8 : 1 (Jun 23)	C_3H_8O
	Cu 3.81 g, O 0.96 g → 0.06 : 0.06 (Jan 23)	CuO
repeat units in a polymer	(242 − 60 for the end groups) ÷ 14 per CH_2 (Nov 24)	13

Titration calculations

METHOD (3–4 MARKS)

1. Moles of the known solution = c × V ÷ 1000. 2. Use the equation ratio. 3. Concentration = moles ÷ volume in dm^3 (or volume = moles ÷ concentration). Give 3 s.f.; never 1 s.f.

Paper	Reaction (ratio)	Data and working	Answer
Jan 20	H_2SO_4 + 2NaOH	0.0250 × 0.200 = 0.00500; ÷ 2; ÷ 0.0167	0.150 mol/dm^3
Jun 21	H_3PO_4 + 3NaOH	30.40 × 0.525 ÷ 1000 = 0.01596; ÷ 3 = 0.00532; ÷ 0.0250	0.213 mol/dm^3
Nov 21	CH_3COOH + KOH (1:1)	45.00 × 0.40 ÷ 1000 = 0.018; ÷ 0.025	0.72 mol/dm^3
Jan 22	$2CH_3COOH$ + $Ba(OH)_2$	0.150 × 22.75 ÷ 1000 = 0.0034125; × 2	0.00683 mol
Jun 22	$2HNO_3$ + $Ba(OH)_2$	21.5 × 0.600 ÷ 1000 = 0.0129; ÷ 2; ÷ 0.025	0.258 mol/dm^3
Jun 23	H_2SO_4 + 2KOH	15.0 × 0.180 ÷ 1000 = 0.00270; × 2; ÷ 0.0250	0.216 mol/dm^3
Nov 23	Na_2CO_3 + $2HNO_3$	25.0 × 0.220 ÷ 1000 = 0.00550; × 2 = 0.0110; ÷ 0.350	31.4 cm^3 of acid
Nov 23	same; 1 mol CO_2 per Na_2CO_3	0.00550 × 24 000	132 cm^3 CO_2
Nov 24	diacid + 2NaOH	0.150 × 0.025 = 0.00375; ÷ 2; ÷ 0.0175	0.107 mol/dm^3
Jun 25	H_2SO_4 + 2KOH	0.025 × 0.500 = 0.0125; ÷ 2 = 0.00625; ÷ 0.0265	0.236 mol/dm^3

The common wrong answers in the mark schemes come from skipping or inverting the ratio (0.516 or 1.032 instead of 0.258; 0.108 or 0.054 instead of 0.216). Write the ratio step on its own line.

11 TITRATION PRACTICAL, SALTS AND SOLUBILITY CURVES

- **Apparatus:** pipette for a fixed volume of alkali (25.0 cm^3), burette for the acid, conical flask, white tile. A pipette is more precise/accurate than a measuring cylinder (Jun 22). A dropping pipette is rejected.
- **Indicators:** methyl orange is yellow in alkali and red in acid; phenolphthalein is pink in alkali and colourless in acid. Universal indicator is unsuitable: no sharp colour change at the end-point.
- **Reasons:** white tile, to see the colour change clearly; swirling, to mix the solutions (Jun 23, Jun 25).
- **Burette readings** to 2 d.p.: titre = final − initial, e.g. 23.80 − 2.15 = 21.65 cm^3 (Nov 24). Concordant titres agree within 0.20 cm^3; tick them and average only those.

MODEL ANSWER · TITRATION METHOD (6) · JAN 23 (ALKALI IN BURETTE); SAME POINTS JUN 26

Rinse the burette with the solution it will hold, then fill it. Make sure the jet below the tap is full (no air bubbles). Use a pipette to transfer 25.0 cm^3 of the other solution to a conical flask. Add a few drops of indicator (not universal indicator). Add the solution from the burette, swirling the flask, until the indicator changes colour permanently; add dropwise near the end-point. Record the initial and final burette readings. Repeat until concordant results are obtained (within 0.20 cm^3).

Faults examiners asked you to spot (Jan 20, Jun 26): flask rinsed with the alkali instead of water; burette rinsed with water instead of the acid; a measuring cylinder instead of a pipette; no initial reading recorded; no white tile; no swirling; a bung on top of the burette; a beaker instead of a conical flask.

Making salts

MODEL ANSWER · SOLUBLE SALT FROM A TITRATION (4–5) · NOV 24, JUN 25

Repeat the titration with the same volumes but without the indicator. Heat the solution to evaporate some of the water until crystals start to form (crystals form on a glass rod dipped in and cooled). Leave to cool so crystals form. Filter off (or decant) the crystals. Dry them between filter papers or in a warm oven/desiccator. Heating to dryness scores 1 at most; a hot oven or Bunsen burner loses the drying mark.

MODEL ANSWER · INSOLUBLE SALT BY PRECIPITATION (3–5) · JAN 20, JAN 21, JAN 23, JUN 24

Dissolve each solid in (distilled) water; mix the two solutions; filter to collect the precipitate; wash it with distilled water; dry it in a warm oven or between filter papers. Any attempt to evaporate the filtrate to form crystals caps the answer at 1–2.

Ionic equation with state symbols (Jan 20): $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$. Insoluble salts that appear: $BaSO_4$, $PbSO_4$, $CaSO_4$, AgCl, $PbBr_2$. All nitrates, and magnesium sulfate, are soluble.

Solubility curves

DEFINITION · SATURATED SOLUTION (2) · NOV 21

A solution containing the maximum mass of solute that can dissolve in the solvent at a particular temperature.

Skill	Example
scale to another mass of water	45 g per 100 g from 4.5 g in 10 g; 124 g per 100 g → 310 g in 250 g (Nov 21); 24 g × 4 = 96 g per 400 g (Jun 24)
mass crystallising on cooling	read both temperatures on the curve and subtract: 21 − 8 = 13 g (Jun 24); 31 − 13 = 18 g (Jan 22). Draw the read-off lines on the graph or lose M1 and M2
scale, then subtract (Jun 26)	12.5 g per 100 cm^3 at 27.4 °C → 12.5 ÷ 4 = 3.13 g in 25 cm^3. 200 cm^3 cooled from 55.6 °C to 27.4 °C: (17.5 − 12.5) × 2 = 10.0 g crystallises
two curves	they have equal solubility where the curves cross (≈ 68 °C, Jun 24)
extrapolate	extend the curve smoothly before reading (0 marks if the curve is not extended, Jun 21)

- Anomalous point reasons (Jun 21): less water used than stated; temperature lower than recorded; not enough solid added; solution not stirred. Results too high (Nov 21): some water evaporated.
- Cooling method (Jun 26): dissolve known masses in 25 cm^3 of water, cool until crystals first appear and record that temperature with a thermometer; the independent variable is the mass of solute. It cannot be used for barium sulfate because barium sulfate is insoluble.
- Practical reasons (Nov 21): heating too quickly could make the solution spit or boil; repeat and take the mean. Hydrated copper(II) sulfate cannot be heated to high temperatures: it loses its water of crystallisation (Jan 22).

12 ENERGETICS AND BOND ENERGIES

- **Exothermic:** heat (thermal) energy is given out to the surroundings. “Energy” alone is not enough.
- In an equation, ⇌ means the reaction is reversible and ΔH is the enthalpy (heat energy) change.
- **Calorimetry:** Q = m c ΔT, where m is the mass of the solution: 25 + 22 = 47 g; Q = 47 × 4.2 × 35 = 6909 J = 6.9 kJ (Nov 23). Combustion of ethanol (Jun 22): 0.92 ÷ 46 = 0.020 mol; ΔH = −18.2 ÷ 0.020 = −910 kJ/mol.
- Errors that make ΔH less exothermic than the data-book value: heat lost to the surroundings; incomplete combustion (black soot is carbon); ethanol evaporating or impure.
- Variables to control in a neutralisation (Nov 23): concentration of the alkali, rate of stirring, starting temperature. A polystyrene cup is a better insulator than glass, so less heat is lost and the temperature

change is more accurate. (“Prevents heat loss” was allowed here, but “reduces” is the safe word.)

MODEL ANSWER · TWO-LINE TEMPERATURE GRAPH (3) · NOV 23

The first line shows the temperature rising as more sodium hydroxide is added, because the neutralisation releases heat. Where the lines cross (the highest temperature) the acid has been completely neutralised. The second line falls because there is no further reaction and the cooler sodium hydroxide solution lowers the temperature.

BOND ENERGY METHOD (3–5 MARKS)

ΔH = Σ(bonds broken in reactants) − Σ(bonds made in products). Draw displayed formulae and count every bond, including coefficients. Negative ΔH = exothermic. In “show that” questions every step must be written.

To find an unknown bond energy, write the equation with x and solve: 4x + 996 − 3462 = −890, so x = 394 (C–H, Jun 22).

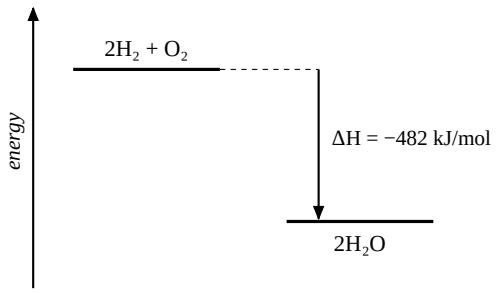
Paper	Reaction	Broken	Made	ΔH / kJ/mol
Jun 19	$N_2H_4 + 2H_2O_2 \rightarrow N_2 + 4H_2O$	3861	4649	−788
Jan 20	$2H_2O_2 \rightarrow 2H_2O + O_2$	2138	2350	−212
Nov 21	$H_2 + F_2 \rightarrow 2HF$	594	1124	−530
Jan 23	$3N_2H_4 \rightarrow 4NH_3 + N_2$	5166	5637	−471
Nov 23	$2H_2 + O_2 \rightarrow 2H_2O$	1370	1852	−482
Jun 24	$H_2 + Cl_2 \rightarrow 2HCl$	678	862	−184
Nov 24	alkene + Br_2 (C=C + Br–Br → C–C + 2 C–Br)	805	900	−95 (show that)
Nov 25	$CH_3Cl + 3Cl_2 \rightarrow CCl_4 + 3HCl$, ΔH = −339	1968 + x	4x + 1293	C–Cl = 338

MODEL ANSWERS · WHY EXOTHERMIC (2) AND WHY ESTERIFICATION HAS ΔH ≈ 0 (2) · JAN 23, NOV 24, JUN 23

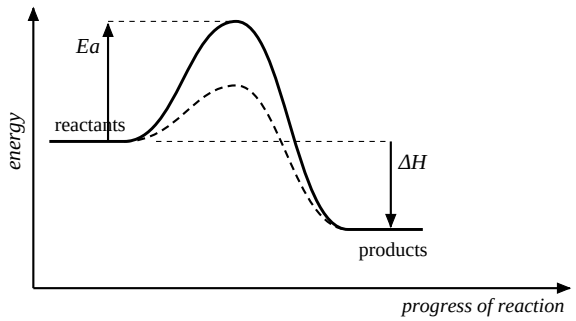
Bond breaking takes in energy and bond making releases energy; more energy is released when the bonds in the products are made than is taken in when the bonds in the reactants are broken.

Esterification: the same bonds (C–O and O–H) are broken and formed, so the energy needed to break bonds equals the energy released when bonds form.

Energy diagrams



(a) energy level diagram, exothermic (Nov 23)



(b) reaction profile; dashed = catalysed (Jun 24, Jun 25)

Energy level diagram (2–3 marks): products line below the reactants for exothermic, both lines labelled with formulae, a vertical ΔH arrow from the reactant level to the product level. Reaction profile (4–5 marks, Jun 24, Jun 25, Nov 25): add the hump, an upward Ea arrow from the reactant level to the peak, and a downward ΔH arrow. A catalysed profile starts and ends at the same levels with a lower peak (Jan 22). If your calculated ΔH was positive, an endothermic diagram earns follow-through marks.

MODEL ANSWER · EFFECT OF INCREASING TEMPERATURE ON RATE (4) · JUN 25

The rate increases. Particles have more kinetic energy and move faster, so there are more collisions per unit time (more frequent collisions), and more of the collisions have energy equal to or greater than the activation energy, so there are more frequent successful collisions. “Rate of collision increases” is ignored.

MODEL ANSWER · COMPARING CATALYSTS FOR H_2O_2 DECOMPOSITION (5) · JAN 22, NOV 25

Add the same volume of hydrogen peroxide to a conical flask and add the same mass of catalyst; keep the temperature constant. Start the timer. Record the volume of gas produced in a given time (or the time to produce a fixed volume). Repeat with each catalyst. Plot the results and compare the gradients: the most effective catalyst gives the fastest rate.

- Sketching curves (Nov 25): a faster reaction starts at the origin, is steeper, and levels off at the same final volume (50 cm^3) if the amount of limiting reactant is unchanged; a slower one is shallower and levels at the same value. Jun 25: smooth curve from the origin that levels off horizontally.
- Rate experiments with Mg and acids (Jan 21): control the volume of acid, temperature, and the mass and surface area of magnesium; insert the bung quickly so as little gas as possible escapes.
- Oxygen relights a glowing splint (MCQ, Jan 20).

Reversible reactions and equilibrium

DEFINITION · DYNAMIC EQUILIBRIUM (2) · JUN 19, JAN 22, JUN 22, JAN 23

The forward and backward reactions are both taking place, at the same rate, so the concentrations of reactants and products remain constant. (Not “equal”; not “at a constant rate”.)

Reaction (paper)	Change	Effect on yield	Reason to write
$C_2H_4 + H_2O \rightleftharpoons C_2H_5OH$, exothermic (Jan 19)	lower temperature	increases	the forward reaction is exothermic
	lower pressure	decreases	more moles of gas on the left
$N_2O_4 \rightleftharpoons 2NO_2$ (Jun 19)	higher pressure	equilibrium moves left	fewer moles of gas on the left
$CH_4 + H_2O \rightleftharpoons CO + 3H_2$, endothermic (Jun 23)	lower temperature	decreases	the forward reaction is endothermic
	lower pressure	increases	2 mol of gas on the left, 4 on the right
$H_2 + I_2 \rightleftharpoons 2HI$, endothermic (Jun 25)	higher temperature	increases	the forward reaction is endothermic
$H_2(g) + X_2(g) \rightleftharpoons 2HX$ (Nov 21, Jun 25)	pressure	no effect	same number of moles of gas on both sides
any equilibrium	add a catalyst	no effect	forward and backward rates increase equally

In Jun 19 the gas mixture looked darker at higher pressure: the concentration of NO_2 increased (its molecules are closer together). “More NO_2 is produced” was rejected.

MODEL ANSWER · RAISING THE TEMPERATURE OF AN EXOTHERMIC EQUILIBRIUM: YIELD AND RATE (5) · JUN 26

Yield: the forward reaction is exothermic, so the backward reaction is endothermic; the equilibrium shifts to the left (reactants) to oppose the temperature rise, so the yield decreases.

Rate: particles have more kinetic energy, so there are more frequent collisions with energy ≥ the activation energy, so the rate increases. Getting either direction wrong caps the answer at 4.

14 CRUDE OIL, ALKENES AND ALCOHOLS

MODEL ANSWER · FRACTIONAL DISTILLATION OF CRUDE OIL (4) · JAN 21, JAN 23, NOV 24, JUN 24

Crude oil is heated until it vaporises. The vapours rise up the fractionating column. The column is hotter at the bottom than at the top (temperature gradient). The vapours condense at their boiling points, so different fractions are collected at different heights. ("Burned" is rejected; "distillation" alone is ignored.)

Fraction	Use (as credited)
refinery gases (A, top)	bottled gas for heating and cooking
gasoline (petrol)	fuel for cars
kerosene	aircraft fuel; fuel for lamps or heating. It must say fuel: "aircraft" alone scores 0 (Jun 25)
diesel; fuel oil	lorries and buses; fuel for ships and power stations
bitumen (F, bottom)	road surfacing (tar), roofing

MODEL ANSWERS · CRACKING (2, 2, 3) · JAN 20, JUN 21, JAN 23, JUN 24, NOV 24

Definition: breaking down long-chain hydrocarbons (alkanes) into shorter-chain molecules. "Polymers" is rejected.

Conditions: silica or alumina catalyst (zeolites accepted) at 600–700 °C.

Reasons: there is a surplus of long-chain fractions and a greater demand for short-chain ones; shorter alkanes are more flammable, so they are more useful as fuels such as petrol; alkenes are produced, which are needed to make polymers (and ethanol).

Balancing: $C_{11}H_{24} \rightarrow C_5H_{12} + C_2H_4 + C_4H_8$ (Jan 20); $C_{12}H_{26} \rightarrow C_9H_{20} + C_3H_6$ (Jun 21).

- **Hydrocarbon:** a compound containing hydrogen and carbon atoms only ("only" carries its own mark; "hydrogen and carbon molecules" is rejected). **Unsaturated:** contains a C=C double bond.
- **Homologous series** (any 2): same general formula; same functional group; consecutive members differ by CH_2; similar chemical properties; trend in physical properties. Alkanes C_nH_{2n+2}.
- **Test for an alkene:** add bromine water; it decolourises (orange → colourless). "Bromine" alone, "clear" or "changes colour" lose the mark; mentioning UV light limits you to 1.

Making ethanol

	Hydration of ethene	Fermentation of glucose
equation	$C_2H_4 + H_2O \rightleftharpoons C_2H_5OH$	$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
conditions	steam; 300 °C (250–350); 60–70 atm; phosphoric acid catalyst	aqueous glucose; enzymes in yeast; 30 °C (20–40); 1 atm; no air
raw material	ethene from crude oil: finite, non-renewable	sugar cane: renewable, but uses land needed for food
rate and process	fast; continuous	slow; batch
product	pure ethanol	impure (aqueous) ethanol, must be distilled

MODEL ANSWERS · FERMENTATION QUESTIONS · JUN 19, JUN 23, NOV 25

Why no air (2): fermentation must be anaerobic; otherwise oxygen oxidises the ethanol to ethanoic acid (or only carbon dioxide and water form).

Why 30 °C (2): below about 30 °C the enzymes are not very active, so fermentation is too slow; above about 40 °C the enzymes in the yeast are denatured.

Compare the two methods (4–6): give linked points from the table, e.g. “fermentation uses a lower temperature and pressure, so less energy is needed”; “hydration is faster and continuous”; “sugar cane takes up land that could grow food”. “Cheaper” on its own is ignored.

- Molecular formula of ethanol when asked: C_2H_6O. Jan 22 rejected C_2H_5OH for a molecular formula.
- Combustion: $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$. A fuel releases heat (thermal) energy when burned.
- Dehydration of ethanol gives ethene (Jan 19). Making ethanol from crude oil is unsustainable because crude oil is finite.
- Oxidation: heat with acidified potassium dichromate(VI) (acidify with sulfuric acid); orange → green; ethanol → ethanoic acid. Alcohols are flammable, so heat in a water bath with no naked flame.
- Name from the formula: $CH_3CH_2CH_2OH$ is propan-1-ol (propan-2-ol rejected, Nov 24).

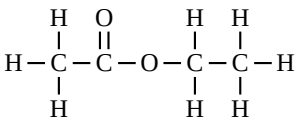
15 CARBOXYLIC ACIDS, ESTERS AND POLYMERS

Reaction of a carboxylic acid	Equation	Observations
with a reactive metal	$2CH_3COOH + Mg \rightarrow (CH_3COO)_2Mg + H_2$; $2CH_3COOH + 2Na \rightarrow 2CH_3COONa + H_2$	fizzing; metal gets smaller
with a carbonate	$2CH_3COOH + Na_2CO_3 \rightarrow 2CH_3COONa + CO_2 + H_2O$	fizzing; solid disappears; vinegar smell fades
methanoic acid	$2HCOOH + Na_2CO_3 \rightarrow 2HCOONa + CO_2 + H_2O$ (NaCOOH rejected)	fizzing
with an alcohol	$CH_3OH + HCOOH \rightleftharpoons HCOOCH_3 + H_2O$ (conc. H_2SO_4 catalyst)	sweet/fruity smell; oily layer

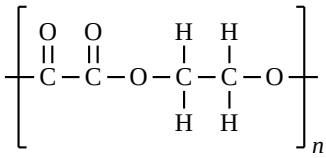
Test for a carboxylic acid (Jan 22): add a carbonate (or Mg, Zn, Fe); effervescence; the gas turns limewater milky. Group 1 metals are rejected.

Esters

- Naming: the alcohol part comes first (-yl), the acid part second (-oate). Ethanol + ethanoic acid → ethyl ethanoate; methanol + ethanoic acid → methyl ethanoate; propanol + methanoic acid → propyl methanoate; butanol + ethanoic acid → butyl ethanoate ($C_6H_{12}O_2$); ethanol is needed to make ethyl propanoate.
- From a formula: $CH_3CH_2COOCH_3$ is methyl propanoate (Nov 23); $CH_3COOCH_2CH_2CH_2CH_3$ is butyl ethanoate (Jun 24). The group bonded to the C=O is the acid part.
- Conditions: concentrated sulfuric acid as catalyst; warm in a water bath because ethanol is flammable. Esterification is reversible; an open container is not a closed system, so equilibrium is not reached (Jun 26).



ethyl ethanoate, $CH_3COOC_2H_5$



polyester repeat unit (Nov 23)

Marks for a displayed ester (2): M1 for the correctly displayed –C(=O)–O– linkage, M2 for the rest, with every C–H and O–H bond drawn.

Condensation polymers

KEY POINTS EXAMINERS CREDIT · JAN 20, NOV 21, JAN 22, JUN 22, NOV 23, NOV 24, NOV 25, JUN 26

A diol + a dicarboxylic acid → a polyester + water. Each ester link releases one H_2O: the –OH comes from the acid and the –H from the diol. For a balanced equation with n, put n before each monomer and 2n H_2O.

Repeat unit (2): M1 for the ester linkage, M2 for the rest with the correct number of carbons from each monomer and extension bonds. Brackets, n and + H_2O are usually ignored.

Addition versus condensation: addition polymerisation uses an alkene monomer and the polymer is the only product; condensation uses two different functional groups (often two monomers) and also produces a small molecule (water).

Polyesters can be biodegradable (broken down by bacteria), unlike poly(alkenes). Landfill: sites fill up, non-biodegradable. Burning: toxic gases or CO_2, a greenhouse gas. “Harmful gases” alone is ignored.

- Jun 26 table: a monomer for an addition polymer = alkene; a monomer for a condensation polymer = dicarboxylic acid (or diol); the type that biodegrades = condensation polymer; a product of condensation polymerisation = water. Succinic acid $C_4H_6O_4$ + 2Na → disodium salt + H_2.
- Addition polymer repeat unit (Jan 19): single C–C bond in the unit, extension bonds through the brackets, n after the bracket.

16 CHEMICAL TESTS USED IN PAPER 2C

Test for	Method	Positive result (rejected wording)
Li^+ / K^+ / Cu^{2+}	flame test	red/crimson / lilac / blue-green (not “brick red”, not “orange-red”)
Cu^{2+} / Fe^{2+}	add sodium hydroxide solution	blue precipitate / green precipitate
Cl^-	add dilute nitric acid, then silver nitrate solution	white precipitate (HCl or H_2SO_4 loses the acid mark)
CO_3^{2-}	add dilute acid; bubble the gas through limewater	effervescence; limewater turns milky
H_2	lighted splint	squeaky pop (glowing splint rejected)
O_2	glowing splint	relights
Cl_2	damp litmus paper	bleached (turns white); blue litmus turns red then white
CO_2	bubble through limewater	milky/cloudy (adding other reagents scores 0)
water	anhydrous copper(II) sulfate	white → blue (cobalt chloride blue → pink)
alkene	bromine water	orange → colourless
purity of a liquid	measure the boiling point	pure substance boils at a fixed temperature (ethanol 78 °C)

Blue litmus in potassium chloride solution is a test for chlorine gas, not chloride ions (Jun 26). The chloride test needs nitric acid and silver nitrate.

17 PRACTICAL AND GRAPH SKILLS

- **Plotting:** every point within ± half a small square. Draw a smooth curve (or two ruled straight lines that meet) through or close to the points. Circle the anomalous point.
- **Reading from a graph:** draw the vertical and horizontal construction lines. Several mark schemes (Jan 23, Jun 24) award the reading marks only if working is shown on the graph.
- **Apparatus names:** thermometer, (Liebig) condenser and beaker in distillation (Nov 24); pipette, burette, conical flask, measuring cylinder, evaporating basin, gas syringe, polystyrene cup.
- **Means:** average the consistent results only, e.g. (64 + 63 + 60) ÷ 3 = 62 (Jun 19); including the anomaly scores 1.
- **Describing a trend:** "as the number of carbon atoms increases, the time taken increases" (Jan 21); "the less easily the alcohol evaporates" (Jun 19). Refer to both variables.
- **Variables:** name them specifically: volume of alcohol, temperature of the water (Jun 19); concentration of alkali, rate of stirring (Nov 23). "Amount" is weaker than "volume" or "mass".
- **Safety:** fume cupboard for chlorine because it is toxic; goggles because acid or bleach is corrosive/irritant (Jun 19); alcohols are flammable, so no naked flame.

MODEL ANSWER · MAKING A STANDARD SOLUTION OF KCL (6) · JUN 26

Moles = 0.500 mol/dm^3 × 0.250 dm^3 = 0.125 mol; mass = 0.125 × 74.5 = 9.31 g. Weigh 9.31 g of KCl on a balance. Transfer it to a beaker (or volumetric flask). Add some distilled water and stir until it dissolves. Make up to a total volume of 250 cm^3 with distilled water (to the line of a volumetric flask) and invert to mix.

18 FINAL A* CHECKLIST

1. Read the command word. Explain = linked reasons with "so" or "because"; Describe = what you do or see; Give/State = a short fact. Count the marks and make that many separate points.
2. Decide the particle that moves before writing about conduction: electrons in metals and graphite, ions in molten or dissolved ionic compounds, nothing in covalent molecules.
3. Name the ion, not the element, in every electrolysis statement: "chloride ions lose electrons"; "Cu^{2+} ions gain electrons".
4. For every yield question: effect + exothermic/endothermic or moles of gas. For every catalyst question: alternative pathway with lower Ea, and no effect on yield.
5. For every calculation: moles first, ratio second, units checked (cm^3 vs dm^3; g vs tonnes), 3 s.f. or standard form as asked, sign on ΔH.
6. For titration data: 2 d.p. readings, concordant titres within 0.20 cm^3, mean to 2 d.p.
7. For displayed formulae: every bond, including O–H; ester link drawn in full; extension bonds on repeat units.
8. For colour changes: both colours, in the correct order (orange → green; yellow → red; orange → colourless).
9. For conditions: number + unit + catalyst (hydration 300 °C, 60–70 atm, H_3PO_4; cracking 600–700 °C, silica/alumina; fermentation 30 °C, yeast, no air).
10. Leave 5 minutes to recheck calculations; most lost A* marks are ratio slips, unit slips and missing signs.